

Readout Frames and Operator Compression: A Modest Note on Structural Coherence, Phase Transport, and Soft Reading

Jan Mikulik

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Abstract

This note isolates a compact structural viewpoint: many apparent paradoxes do not belong to the source object, but to the chosen readout frame. The exact technical core is an operator-level compression of the chain rule, together with a small collection of elementary readout lemmas. Around this core we place two carefully labeled additional layers: a geometric ansatz for phase/intensity separation and an interpretive soft-reading principle describing survival of structure under degraded observation.

The purpose of the note is not to claim that classical theorems are new, but to show that a single readout-oriented language usefully organizes several familiar phenomena: substitution, Euler's formula, projection paradoxes, local phase winding, and the distinction between hard and soft observation.

Guiding principle. Wrong frame $\Rightarrow$ fake magic. Right frame $\Rightarrow$ clean structure.

1 The exact operator core

1.1 Substitution and transport channels

Let $u = u(x)$ be a differentiable function. Define the substitution operator U_u and the multiplication operator $M_{u'}$ by

$$U_u[F](x) := F(u(x)), \quad M_{u'}[g](x) := u'(x) g(x).$$

Definition 1.1 (Compressed transport operator). *The CORE operator associated with the channel u is*

$$\text{CORE}_u := M_{u'} \circ U_u.$$

The point of this notation is not to replace the chain rule, but to compress its two-step form into a single transport object.

Proposition 1.2 (Compressed chain rule). *For every differentiable F one has*

$$D_x \circ U_u = \text{CORE}_u \circ D_u,$$

equivalently,

$$\frac{d}{dx} F(u(x)) = u'(x) F'(u(x)).$$

Proof. By definition,

$$(D_x \circ U_u)[F](x) = \frac{d}{dx} F(u(x)).$$

On the other hand,

$$(\text{CORE}_u \circ D_u)[F](x) = (M_{u'} \circ U_u)[F'](x) = u'(x) F'(u(x)).$$

These are equal by the classical chain rule. $\square$

Remark 1.3. *The novelty claimed here is not the underlying calculus identity, but its operator-level packaging into a single readout step generated by CORE_u .*

1.2 Antiderivative form

The same compression appears in substitution for integrals.

Proposition 1.4 (Compressed antiderivative transport). *If $F'(u) = f(u)$, then*

$$\int f(u(x)) u'(x) \, dx = F(u(x)) + C.$$

Equivalently,

$$\int \text{CORE}_u[f](x) \, dx = F(u(x)) + C.$$

Proof. This is the usual substitution rule written in the present notation. $\square$

2 Euler's formula as readout of a real rotation

We do not claim Euler's formula to be new. The point is that it may be read as a real rotation viewed through the correct output map.

Consider the real rotation generator

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

on $\mathbb{R}^2$, and define the complex readout map

$$\Pi : \mathbb{R}^2 \rightarrow \mathbb{C}, \quad \Pi(x, y) := x + iy.$$

Proposition 2.1 (Intertwining relation). *The map Π satisfies*

$$\Pi \circ J = i \Pi.$$

Proof. For $(x, y) \in \mathbb{R}^2$,

$$J(x, y) = (-y, x),$$

hence

$$(\Pi \circ J)(x, y) = \Pi(-y, x) = -y + ix = i(x + iy) = i \Pi(x, y).$$

$\square$

Corollary 2.2 (Exponential transport of rotation). *For every $\theta \in \mathbb{R}$,*

$$\Pi \circ e^{\theta J} = e^{i\theta} \Pi.$$

Proof. Exponentiate the intertwining relation. $\square$

Apply both sides to the basis vector $e_1 = (1, 0)$. Since

$$e^{\theta J} e_1 = (\cos \theta, \sin \theta),$$

one obtains

$$e^{i\theta} = \Pi(e^{\theta J} e_1) = \Pi(\cos \theta, \sin \theta) = \cos \theta + i \sin \theta.$$

Remark 2.3. *In this frame, Euler's formula is not magic. It is the complex readout of a real rotation. The phase is geometry remembered by the operator.*

3 A hidden lemma on readout contradiction

The following elementary statement carries much of the philosophical weight of the readout-frame viewpoint.

Lemma 3.1 (Contradiction may belong to the readout). *Let V and W be vector spaces, and let $\{P_\theta : V \rightarrow W\}_{\theta \in \Theta}$ be a family of readout maps. Assume the family is nontrivial, i.e. there exist $\theta_1 \neq \theta_2$ such that*

$$P_{\theta_1} \neq P_{\theta_2}.$$

Then there exists an object $X \in V$ such that

$$P_{\theta_1}(X) \neq P_{\theta_2}(X), \quad \text{while of course} \quad X = X.$$

Proof. Since $P_{\theta_1} \neq P_{\theta_2}$, the difference

$$P_{\theta_1} - P_{\theta_2} : V \rightarrow W$$

is not the zero map. Hence there exists $X \in V$ such that

$$(P_{\theta_1} - P_{\theta_2})(X) \neq 0.$$

Equivalently,

$$P_{\theta_1}(X) \neq P_{\theta_2}(X).$$

This concerns two outputs of the same object X under two different readouts. It does not alter the identity of the source. $\square$

Corollary 3.2 (Projection paradox). *A single rigid source may admit mutually incompatible visible appearances under different readout frames:*

$$P_{\theta_1}(X) \neq P_{\theta_2}(X) \quad \text{although} \quad X = X.$$

Thus projection can create contradiction in appearance without contradiction in the source.

Remark 3.3. *This is the clean abstract form of the slogan: the contradiction may belong to the readout, not to the object.*

4 A fourfold readout ansatz

The present section is *not* a theorem about the zeta function, quantum mechanics, or reality in general. It is a structural ansatz designed to separate visible amplitude from hidden phase.

Definition 4.1 (Fourfold readout ansatz). *Let*

$$\psi(r, \theta) = R(r)(a + b \cos(4\theta))e^{i\Phi(r, \theta)},$$

where $R(r) \geq 0$, $a, b \in \mathbb{R}$, and $\Phi(r, \theta)$ is a phase field. Define the observable intensity by

$$I(r, \theta) := |\psi(r, \theta)|^2.$$

Since $|e^{i\Phi}| = 1$, one gets

$$I(r, \theta) = |R(r)|^2 (a + b \cos(4\theta))^2.$$

Observation 4.2 (Separation of layers). *The ansatz separates two different layers:*

(i) a visible amplitude layer, governed by

$$|R(r)|^2 (a + b \cos(4\theta))^2,$$

(ii) a hidden transport layer, governed by

$$e^{i\Phi(r, \theta)}.$$

The phase influences the state, but not directly the intensity modulus.

Observation 4.3 (Fourfold symmetry). *The intensity is invariant under the quarter-turn*

$$\theta \mapsto \theta + \frac{\pi}{2},$$

because

$$\cos\left(4\left(\theta + \frac{\pi}{2}\right)\right) = \cos(4\theta + 2\pi) = \cos(4\theta).$$

Hence

$$I\left(r, \theta + \frac{\pi}{2}\right) = I(r, \theta).$$

Remark 4.4. *This ansatz is best read as a phase/readout toy model: hidden transport may coexist with a visible symmetry pattern without being directly legible in the intensity alone.*

5 Local phase winding and critical-line readout

This section separates an exact local fact from a weaker one-dimensional readout.

Theorem 5.1 (Simple zero gives a 2π local winding). *Let f be holomorphic near $s_0 \in \mathbb{C}$, and suppose s_0 is a simple zero of f . Then for every sufficiently small positively oriented circle*

$$\gamma_r(\phi) = s_0 + re^{i\phi}, \quad 0 \leq \phi \leq 2\pi,$$

one has

$$\Delta_{\gamma_r} \arg f = 2\pi.$$

Proof. Since s_0 is a simple zero,

$$f(s) = (s - s_0)g(s),$$

where g is holomorphic and nonzero near s_0 . Along the small circle,

$$s - s_0 = re^{i\phi},$$

so the factor $(s - s_0)$ contributes exactly 2π to the argument variation. The factor g contributes 0 for sufficiently small r , since it has no zeros there. Hence the total variation is 2π . $\square$

Remark 5.2. *Thus a simple zero carries a local topological winding number 1.*

Now specialize to the critical line and Hardy's real readout

$$Z(t) := e^{i\theta(t)} \zeta\left(\frac{1}{2} + it\right),$$

where $\theta(t)$ is chosen so that $Z(t) \in \mathbb{R}$ for real t .

Remark 5.3 (Two readouts of the same local structure). *A simple zero on the critical line admits two natural local readouts:*

(i) *in the complex local frame, it appears as a full 2π argument winding,*

(ii) *in the real Hardy frame, it appears only as a sign flip of $Z(t)$.*

The source structure is the same; the visible manifestation depends on the readout frame.

6 Soft reading as structural extension

Everything up to this point has been exact and elementary. We now introduce a separate interpretive layer. It is not a replacement for the exact results above, but an attempt to formalize the intuition that direct orientation may fail while a weaker readable channel survives.

Definition 6.1 (Soft reading family). *Let ρ be a state and let $\{M_k^{(\eta)}\}_k$ be a family of measurement operators with reading strength $\eta > 0$ such that*

$$\sum_k (M_k^{(\eta)})^\dagger M_k^{(\eta)} = I.$$

We call the family a soft-reading family if the disturbance satisfies

$$D_\eta(\rho) \rightarrow 0 \quad (\eta \rightarrow 0),$$

while the information yield remains nonzero for every $\eta > 0$:

$$I_\eta > 0.$$

Assume that the observable output does not couple directly to an unstable oriented quantity ξ , but rather to the regularized magnitude

$$r_\varepsilon(\xi) := \sqrt{\xi^2 + \varepsilon^2}.$$

Suppose the effective detection functional admits an expansion

$$\Pi_{\eta,\varepsilon}(\xi) = \Pi_0 + \eta W(r_\varepsilon(\xi)) + O(\eta^2),$$

where $W \geq 0$ and $W \not\equiv 0$.

Corollary 6.2 (Nonzero soft-reading chance). *If*

$$W(r_\varepsilon(\xi)) > 0,$$

then for sufficiently small $\eta > 0$ one has

$$\Pi_{\eta,\varepsilon}(\xi) > \Pi_0.$$

Proof. The first-order term in η is strictly positive, so the $O(\eta^2)$ term cannot cancel it for sufficiently small positive η . $\square$

Remark 6.3 (From soft readability to branch legality). *The previous corollary says: loss of direct sign does not imply loss of observability. A weaker readable channel may survive through an effective magnitude law.*

If one interprets the excess

$$\Gamma_{ij} := \Pi_{\eta,\varepsilon}(\xi) - \Pi_0$$

as an admissible transition rate, then

$$W(r_\varepsilon(\xi)) > 0 \implies \Gamma_{ij} > 0.$$

Adding a branch-selection condition

$$|T_j| > |T_i|$$

produces the shorthand notion we will call *luck*: a nonzero legal transition into a better branch.

When sign does not survive, norm may survive; when hard readout fails, soft chance may remain.

Luck is the nonzero legality of a better branch.

7 What is exact, what is model, and what is interpretive

To avoid confusion, we summarize the status of the main ingredients.

- (1) **Exact.** The operator identity

$$D_x \circ U_u = \text{CORE}_u \circ D_u,$$

the Euler readout relation

$$\Pi \circ J = i\Pi,$$

the hidden lemma on readout contradiction, and the local 2π winding of a simple zero are exact as stated.

- (2) **Model / ansatz.** The fourfold polar expression

$$\psi(r, \theta) = R(r)(a + b \cos(4\theta))e^{i\Phi(r, \theta)}$$

is a structural ansatz separating visible intensity from hidden phase.

- (3) **Interpretive extension.** The soft-reading framework and the branch-level language of luck are meant as structural extensions of the readout viewpoint, not as replacements for the exact statements proved earlier.
- (4) **Heuristic caution.** Any strong global claim—for example a full identification of all quantum collapse, all zeta geometry, or all apparent paradox with a single formal theorem—would require additional work and is not claimed here.

8 Conclusion

The present note proposes a modest but useful synthesis.

At the exact level, substitution may be compressed into a single transport operator, real rotation may be read through a complex output map, and apparent contradiction may be located in the readout rather than in the source. At the model level, visible amplitude and hidden phase may be separated. At the interpretive level, loss of direct orientation need not mean loss of readable structure.

The common thread is simple:

$$\text{same source} \not\Rightarrow \text{same appearance under every frame.}$$

Once this is kept in view, many objects that first look mystical become structurally ordinary.

$$\text{Wrong frame} \Rightarrow \text{fake magic.} \qquad \text{Right frame} \Rightarrow \text{clean structure.}$$